

- ◆ You can check complete details of the project at: <http://dgit.in/turnsignaljacket>.

Persistence of Vision wand

This project exploits the phenomenon of ‘Persistence of Vision’.

Persistence of Vision is the theory that an image is thought to persist for approximately one twenty-fifth of a second on the retina of the eye, and believed to be the explanation for motion perception.

A single strip of LEDs was used in the image shown here to create an effect portraying different words. To make a persistence of vision wand like this, you’ll need the following components:

- ◆ Arduino™ Uno
- ◆ 20 LEDs
- ◆ Arduino™ Proto shield
- ◆ 9V alkaline battery
- ◆ 9V battery connector
- ◆ 20 resistors (any value between 100 ohm and 470 ohm)



Persistence of Vision wand in action

How to implement:

- ◆ Build a support to place 20 LEDs one below the other.
- ◆ Connect the resistors along with the LED and connect the anode of every LED to the data pins on Arduino™ board.
- ◆ The resistors are used to limit the current through the LED to avoid damage.
- ◆ Use the proto board to wire the LED to the Arduino™ board.
- ◆ Burn the code to Arduino™ board, switch off the light and test your wand.
- ◆ To change the text of the wand, find the variable ‘povtext’ in the code and put your message there.

You can check the project at <http://dgit.in/LeVision>.

Quadcopter

The movie ‘3 Idiots’ features a flying machine with four motors that allow it to be flown around in any direction as well as hover at a place like a helicopter. The device in this project, however, is much more flexible and